

Security+ 4.5 · Operating System Security

Master CyberSec — Official CompTIA Security+ (SY0-701) Comprehensive Study Guide · **Lesson 4.5** —
Operating System Security (3:22) · [Official YouTube Video ↗](#) · [Professor Messer Study Hub ↗](#)

Executive Lesson Summary: Operating system security includes Active Directory (AD) for centralized identity management in Windows (accounts, groups, GPOs), Group Policy for mass policy deployment, SELinux (MAC in Linux, enforcing least privilege per process), patch baselines, and disabling orphaned accounts.

High-Yield Core Concepts Tested on Exam

- Active Directory is the centralized directory for Windows environments
- SELinux provides Mandatory Access Control (MAC) in Linux, stricter than root
- Group Policy allows mass deployment of security policies
- Patch baseline ensures systems are updated to a known secure state
- Disable orphaned accounts to reduce attack surface
- AD manages accounts, groups, and GPOs for Windows domains

Exam Tips & Classic Scenarios

- If the question mentions 'centralized identity management in Windows', the answer is Active Directory
- SELinux is an example of MAC; it enforces least privilege even for root

Typical Exam Scenario: A company controls access to resources by assigning users to groups in Active Directory. This is an example of centralized identity management using AD.

Beyond the Video: Official CompTIA Exam Concepts, Protocols & Traps (Lesson 4.5 — Operating System Security)

Why is this advanced coverage required for SY0-701? Video overviews provide foundational familiarity, but CompTIA tests practical engineering judgment and requires deep familiarity with the following protocols, acronyms, and operational nuances in this domain:

- **MAC:** Mandatory Access Control: Rigid label-based access architecture (Top Secret, Secret, Unclassified) enforced centrally by security kernel according to formal classification policies.

Classic CompTIA Traps & Question Framing:

- Avoid rote memorization: CompTIA exam prompts regularly adjust technical constraints requiring you to defend why one valid security control is the *MOST appropriate* or *FIRST step*.
- Scrutinize question qualifiers: 'without additional licensing costs' favors native OS tools; 'granular administrative command accounting' favors TACACS+ over RADIUS; 'non-repudiation' mandates asymmetric private key signing.

Practice Questions with In-Depth Solutions (10 Questions)

Detailed answer key with step-by-step technical rationale and comprehensive distractor breakdown provided below. Attempt answering independently first.

1. Schedules a task to run automatically at a fixed time:

- A)** cron (crontab)
- B)** at
- C)** sleep
- D)** nohup

2. Giving each user ONLY the minimum access they need for their role is:

- A)** Least privilege
- B)** Separation of duties
- C)** Job rotation
- D)** Geographic need-to-know

3. Corporate scenario (Operating System Security): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.5), an analyst evaluates the following technical requirement: 'Active Directory: the corporate Windows directory'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Implement the corresponding solution: Active Directory: the corporate Windows directory...
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators with generic credentials

4. Corporate scenario (Operating System Security): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.5), an analyst evaluates the following technical requirement: 'SELinux: MAC on Linux (stricter than root)'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Allow shared access among administrators with generic credentials
- D)** Implement the corresponding solution: SELinux: MAC on Linux (stricter than root)...

5. Corporate scenario (Operating System Security): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.5), an analyst evaluates the following technical requirement: 'centralized identity management in Windows'=Active Directory'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Allow shared access among administrators with generic credentials
- D)** Implement the corresponding solution: 'centralized identity management in Windows'=Active Directory...

6. Corporate scenario (Operating System Security): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.5), an analyst evaluates the following technical requirement: "company controls who has access where via AD groups'=Active Directory.'. What is the best recommendation or applicable control classification?

- A)** Implement the corresponding solution: 'company controls who has access where via AD groups'=Active Directory....
- B)** Temporarily disable audit logs to accelerate system throughput
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

7. Corporate scenario (Operating System Security): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.5), an analyst evaluates the following technical requirement: 'Active Directory: the corporate Windows directory'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Implement the corresponding solution: Active Directory: the corporate Windows directory...
- D)** Allow shared access among administrators with generic credentials

8. Corporate scenario (Operating System Security): During a security review and compliance audit aligned with the CompTIA Security+ (SY0-701, Section 4.5) exam, an analyst evaluates the following technical requirement: 'SELinux: MAC on Linux (stricter than root)'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Allow shared access among administrators with generic credentials
- D)** Implement the corresponding solution: SELinux: MAC on Linux (stricter than root).

9. Corporate scenario (Operating System Security): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.5), an analyst evaluates the following technical requirement: 'centralized identity management in Windows' = Active Directory. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Allow shared access among administrators using generic credentials
- D)** Implement the corresponding solution: 'centralized identity management in Windows' = Active Directory...

10. Corporate scenario (Operating System Security): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.5), an analyst evaluates the following technical requirement: 'company controls who has access where via AD groups'=Active Directory.'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Allow shared access among administrators with generic credentials
- D)** Implement the corresponding solution: 'company controls who has access where via AD groups'=Active Directory

COMPREHENSIVE ANSWER KEY & DISTRACTOR BREAKDOWN

1) A — cron (crontab).

✓ **Why it is correct:** cron (and its interface crontab) is the standard task scheduler in Unix/Linux systems to execute commands or scripts at fixed times, specific dates, or regular intervals defined by cron expressions (minute, hour, day of month, month, day of week). The crontab file allows configuring recurring tasks, such as daily backups, log rotation, or periodic checks. The question asks to schedule a task to run automatically at a fixed time, and cron is exactly the tool designed for recurring time-based scheduling.

✗ **Why the others are wrong:**

- *at*: *at* is a command to schedule a one-time execution of a task at a specific moment in the future (e.g., 'at 15:00' runs once at 3 PM). It is not designed for recurring tasks at a fixed time; after execution, the task is removed from the queue. Although it can be used for one-off scheduling, it does not meet the requirement of running automatically at a fixed time continuously, which is the typical case for periodic scheduling. cron is the correct tool for recurrence.
- *sleep*: *sleep* is a command that suspends the execution of a process for a specified time interval (e.g., 'sleep 60' pauses for 60 seconds). It does not schedule tasks; it only inserts a delay within a script or terminal session. There is no automatic scheduling mechanism at a fixed time, nor persistence across reboots. Therefore, it is completely unsuitable for scheduling automatic tasks at predefined times.
- *nohup*: *nohup* is a command that allows running another command immune to hangup signals (SIGHUP), ensuring the process continues running even after the terminal is closed or the user logs out. It does not schedule tasks; it only protects a running process. It has no functionality for scheduling based on a fixed time. Therefore, it does not meet the requirement to automatically schedule a task to run at a predefined time.

2) A — Least privilege.

✓ **Why it is correct:** The correct answer is 'Least privilege'. The principle of least privilege determines that each user, process, or system should be granted only the permissions strictly necessary to perform its functions, and nothing more. In the information security context, this reduces the attack surface and limits the impact of a compromise. For example, a user in the finance department should not have access to human resources files. CompTIA Security+ SY0-701, in the Architecture and Design domain (2.1), lists least privilege as a fundamental security principle. The question describes exactly this definition: giving each user only the minimum access they need for their role.

✗ **Why the others are wrong:**

- *Separation of duties*: Separation of duties is a control that divides a critical task into steps performed by different people, so that no single person has total control over the process. For example, the person who approves a payment cannot be the same person who executes it. This prevents fraud and errors, but it does not refer to granting minimum access per user; instead, it distributes responsibilities. The question is about individual minimum access, not task division.
- *Job rotation*: Job rotation is the practice of periodically moving employees between different roles to detect fraud, reduce monotony, and train staff. It has no relation to the principle of granting only necessary access. Job rotation is an internal risk mitigation strategy, but it does not define each user's privilege level.
- *Geographic need-to-know*: Need-to-know is a security principle that restricts access to information only to those who need it to perform a specific task. However, 'geographic need-to-know' is not a standard term in the CompTIA Security+ certification. The concept of need-to-know

is often associated with least privilege, but the expression 'geographic' adds a location restriction that is not present in the classic definition. The question does not mention geographic restriction, only functional minimum access. Therefore, the alternative is imprecise and does not correspond to the correct technical term.

3) B — Implement the corresponding solution: Active Directory: the corporate Windows directory....

✓ **Why it's correct:** According to the official CompTIA Security+ exam objectives (SY0-701) for lesson 4.5 (Operating System Security): Active Directory: the corporate Windows directory.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. It makes it impossible to prove who made critical changes or malicious commands.

4) D — Implement the corresponding solution: SELinux: MAC on Linux (stricter than root)....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.5 (Operating System Security): SELinux: MAC on Linux (stricter than root). SELinux (Security-Enhanced Linux) implements Mandatory Access Control (MAC), where the operating system enforces access policies that even the root user cannot override, providing granular control over processes and files.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to incident detection and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls that rely solely on user behavior and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

5) D — Implement the corresponding solution: 'centralized identity management in Windows'=Active Directory....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.5 (Operating System Security): 'centralized identity management in Windows'=Active Directory. Active Directory is Microsoft's directory service that provides centralized authentication, authorization, and management of users, computers, and resources in a Windows domain environment.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and violates compliance standards (PCI DSS, ISO 27001).

- *Replace preventive technical controls with informal terms of adherence*: Informal terms are administrative controls that rely solely on user behavior and do not block automated attacks, injections, or runtime malware.
- *Allow shared access among administrators with generic credentials*: Destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

6) A — Implement the corresponding solution: 'company controls who has access where via AD groups'=Active Directory....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.5 (Operating System Security): 'company controls who has access where via AD groups'=Active Directory. Active Directory is a directory service that centralizes user and resource management, allowing administrators to control access through group memberships and apply security policies across the domain.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput*: Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and violates compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence*: Informal terms are administrative controls that rely solely on user conduct and do not block automated attacks, injections, or runtime malware.
- *Allow shared access among administrators using generic credentials*: Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

7) C — Implement the corresponding solution: Active Directory: the corporate Windows directory....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.5 (Operating System Security): Active Directory: the corporate Windows directory.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput*: Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence*: Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials*: Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

8) D — Implement the corresponding solution: SELinux: MAC on Linux (stricter than root)..

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.5 (Operating System Security): SELinux: MAC on Linux (stricter than root). SELinux implements Mandatory Access Control (MAC), which enforces system-wide policies that even root cannot bypass, providing granular access control beyond traditional discretionary access control.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput*: Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and violates

compliance standards (PCI DSS, ISO 27001).

- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user behavior, and do not block automated attacks, injections, or runtime malware.
- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

9) D — Implement the corresponding solution: 'centralized identity management in Windows' = Active Directory....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.5 (Operating System Security): 'centralized identity management in Windows' = Active Directory.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

10) D — Implement the corresponding solution: 'company controls who has access where via AD groups'=Active Directory.

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.5 (Operating System Security): 'company controls who has access where via AD groups'=Active Directory.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
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